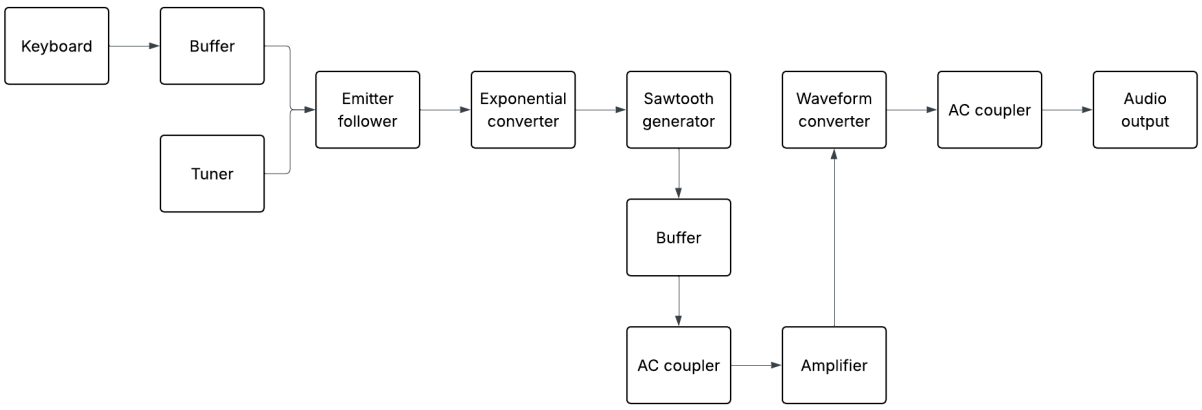
ES 50 Final Project Writeup

EuroRackingIt - Team 31

Henry Aronofsky
Evan Eiling

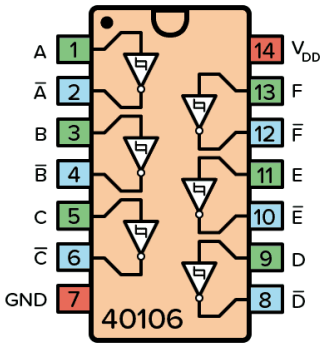
1.a Project Description

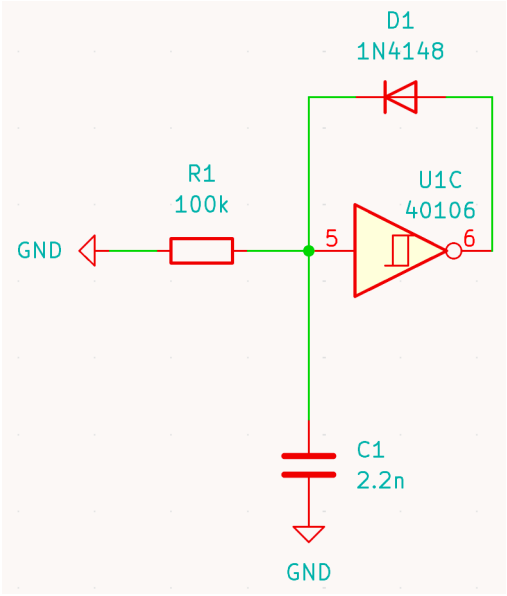
In this project we set out to design and construct a fully analog 12 note piano. To achieve this goal we needed to develop both a keyboard interface which would output a variable voltage depending on the desired note, and a sound generator which could be tuned to the 12 tones equally tempered system and then processed to create a tonally musical output. For the former we employed a 12 step voltage divider. For each resistor node, a button bridged from it to a bus. For the sound generator, we needed two components: a tuning circuit and an oscillator core. The former was composed of a group of voltage adders and dividers to dial in the exact distance between notes. The latter employed a schmitt trigger inverter to create and process an oscillation. The following sections will expound on these circuits. The description will begin at the sawtooth generator and work forwards to the audio output. It will then jump back to the exponential converter and emitter follower and then back again to the keyboard and work its way forward until all components are explained in depth.



2.a Sawtooth generator

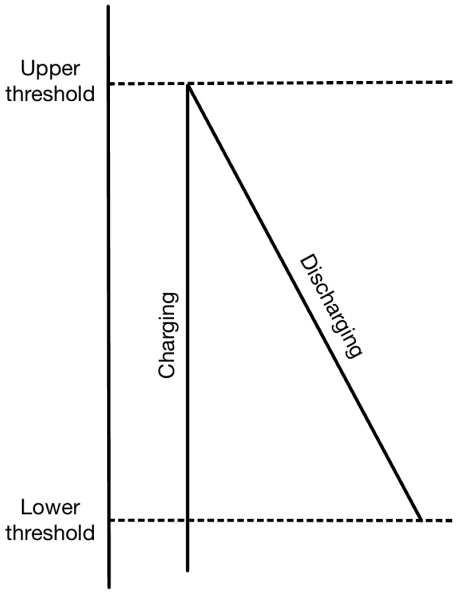
The most crucial component of this project is the oscillator core which ran on a schmitt trigger inverter. This design employed the 40106 which has six such circuits. The schmitt trigger along with a diode, a resistor, and a capacitor were assembled into a basic oscillator.





"IC 40106 (Hex Schmitt Trigger)"
Build Electronic Circuits

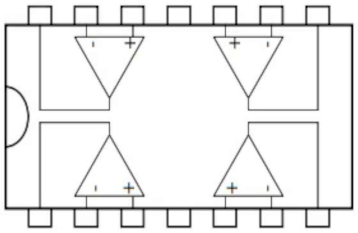
Schmitt trigger inverters act similarly to NOT logic gates but for signals rather than booleans. The schmitt trigger inverter has lower and upper thresholds. When its input node is below its lower threshold the trigger sends its output high. When its input node rises above the upper threshold, it sends its input low.



When this circuit is first powered, the input node is low so the output of the trigger is sent high. During this time, the capacitor begins to charge. The capacitor changes almost instantly causing the input node to exceed the upper threshold of the trigger. The output is sent low, allowing the capacitor to gradually drain to ground, limited in speed by the resistor. Once the capacitor has emptied, the input node is low again and the cycle repeats at a frequency determined by the resistor. The diode is to prevent electrical flow into the output of the trigger.

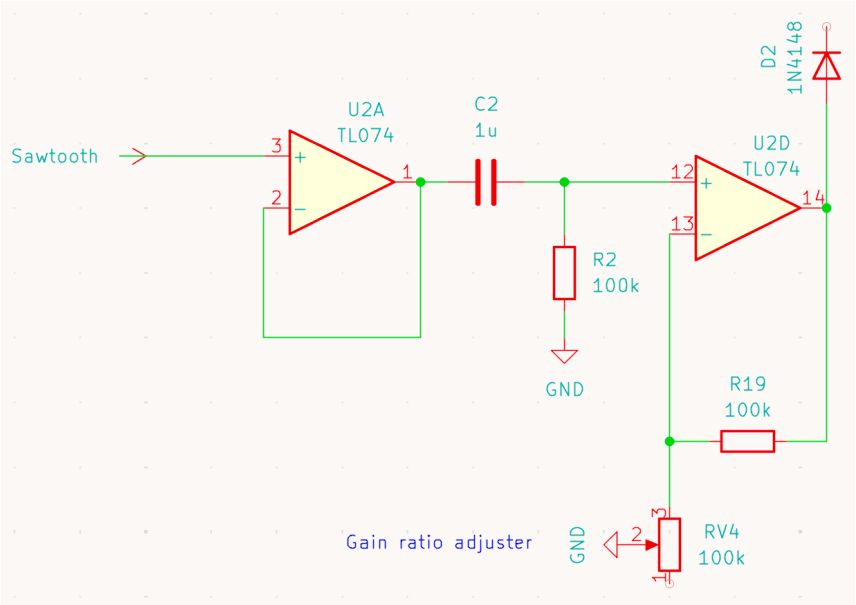
2.b Buffer, AC coupler, Amplifier

For audio processing and amplification this project employed a TL074 IC. This IC is a four in one operational amplifier (op amp) which is used as both a buffer and an amplifier.



"TL074 Op-Amp Pinout…", Microcontrollerslab

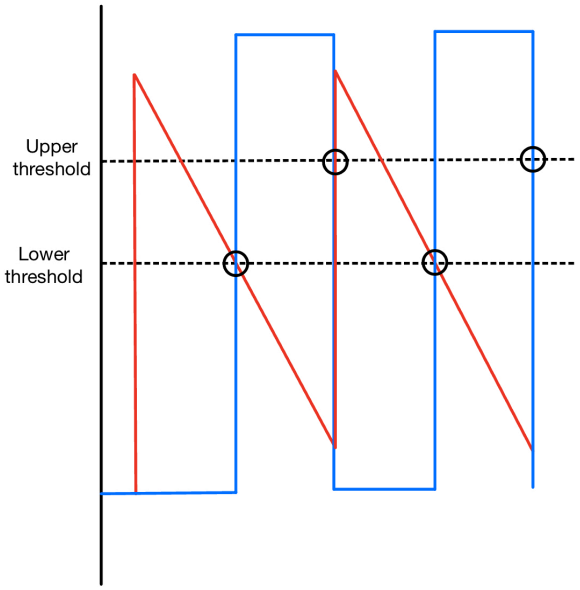
The sawtooth signal generated in the previous step will be offset and not centered around zero volts. This is suboptimal for processing and output as it may exceed the rails of opamps if amplified or damage speaker gear. To prevent this the signal is first fed through an op amp (U2A) in buffer configuration to prevent processing from disrupting or shorting the wave. The output of the buffer is sent through an AC coupler to center the signal at zero volts. An AC coupler works by sending a voltage first through a capacitor which copies the signal and then through a resistor to ground which effectively removes the offset voltage. The now centered signal is safe for amplification and output. To this end, another op amp (U2D) in the same chip is set up in non-inverting amplification configuration with a potentiometer acting as the ground resistor. This allowed for fine adjustment of the gain ratio. A diode at the output prevents electrical back flow from the next stage.



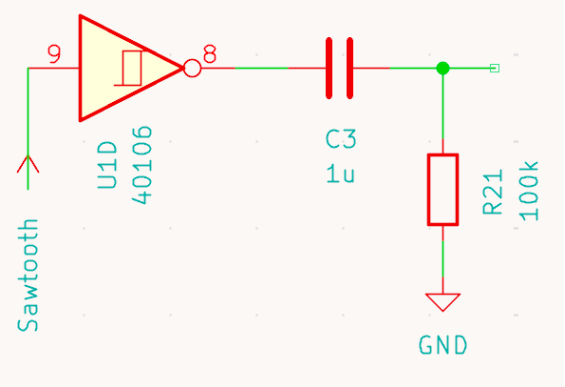
2.c Waveform converter, AC coupler

While the sawtooth wave is musically useful, it lacks any way of adjusting tone as changing the pulse width of the wave affects the pitch as well. In order to develop a more musically useful synthesizer the sawtooth must be converted into a square wave which can be pulse modulated to change tone. Luckily, converting analog signals into digital ones is the primary function of schmitt trigger inverters.

The circuitry for this converter is elementary. The output from the previous amplification stage is fed into the input of another trigger in the 40106 IC and the output can be directly measured as a square wave. This conversion process is best understood by the waveform graph below. The red sawtooth wave is the input signal, while the blue square wave is the output signal. The black circles indicate the moments at which the trigger flips its output as the sawtooth wave crosses its thresholds. As the capacitor discharges, the signal slopes down until it crosses its lower threshold sending the output high and maintaining it high until the sawtooth is sent high, causing the trigger to send its output low. Thus the pulse width of the square wave can be adjusted depending on the amplification of the sawtooth.

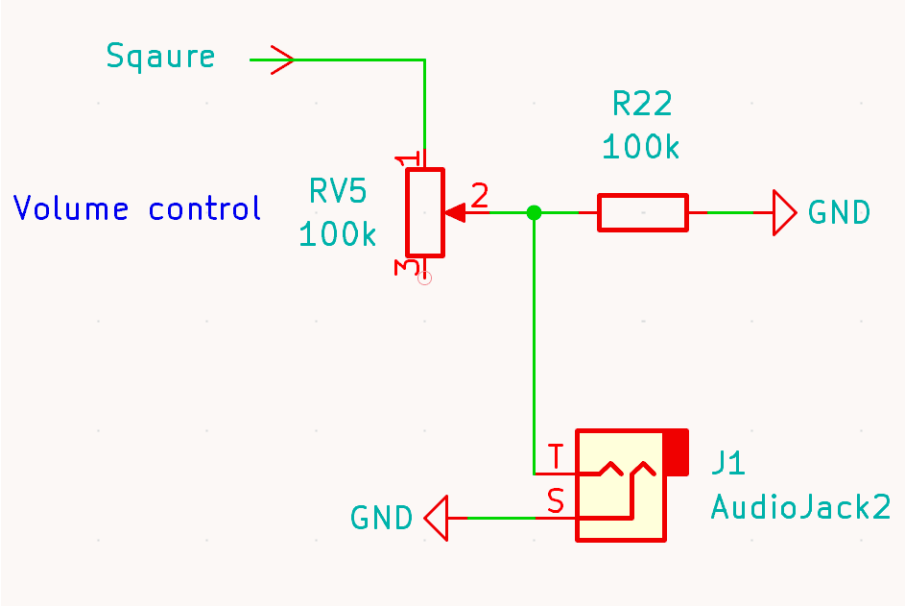


This signal is then sent through another AC coupler in order to center it around zero volts at which point is ready to be listened to.



2.d Audio output

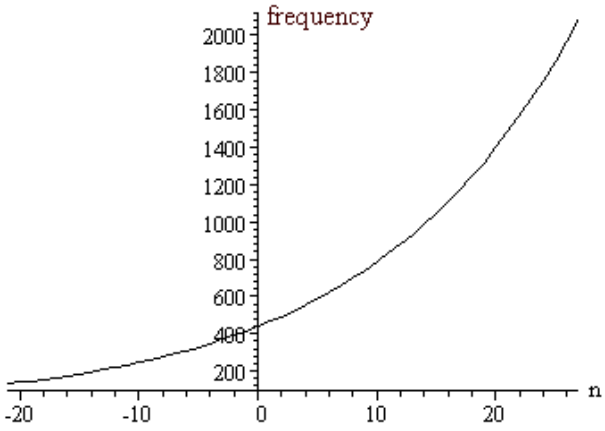
This final stage of the sound generator is also the simplest. A simple voltage divider with a potentiometer is used as a volume control and then the signal is routed through an audio jack and into a pair of headphones.



2.e Exponential converter

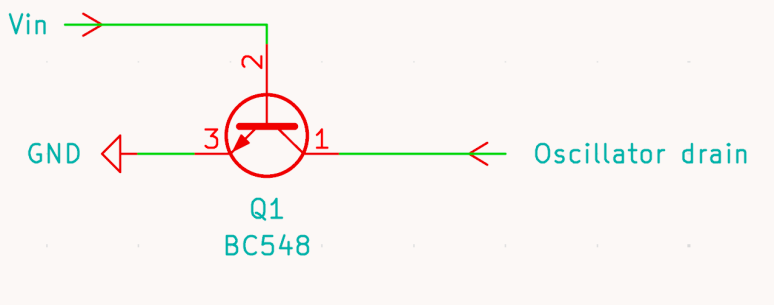
To expound on the tuning circuit it is important to first clarify a previously simplified design attribute of the sawtooth generator. After the capacitor in the sawtooth generator circuit charges and the capacitor fills, the voltage must drain for the cycle to repeat and for there to be oscillations. If a static resistor is used then only one pitch will be accessible by the synthesizer. It is possible to use a potentiometer to vary the frequency over a wide range but in order to create a musically useful device, an exponential converter must be employed.

Musical notes are not organized linearly but rather according to an exponential sequence with each octave doubling in frequency from the previous one. In order to achieve this behaviour, an NPN transistor can be used in its exponential region of operation.



"Music and Trigonometric Graphs", IntMath

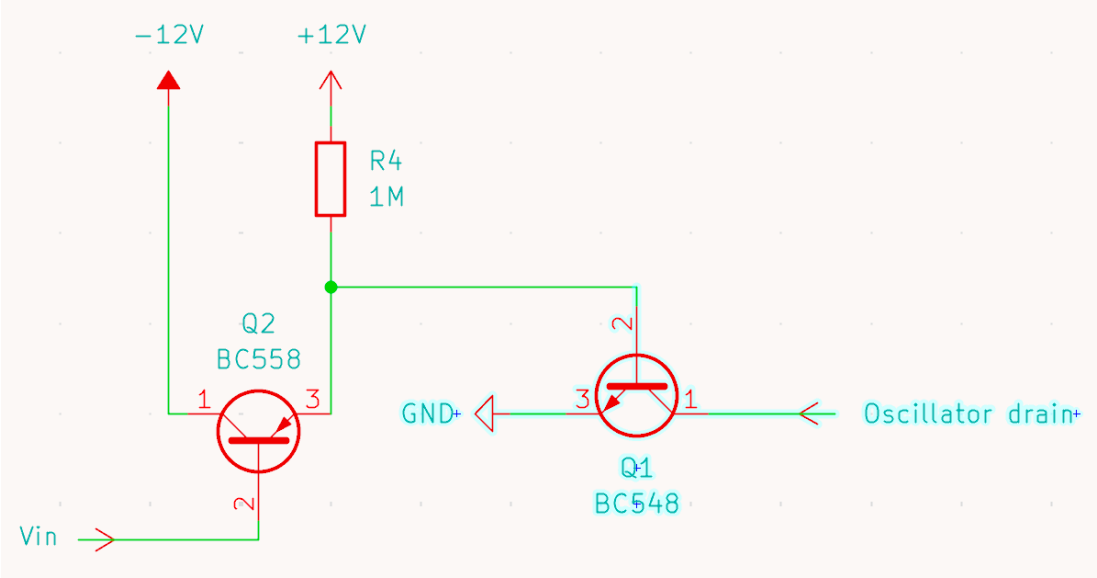
NPN transistors open when a voltage is applied to their base. Above a volt they operate in a linear region which makes the transistor operate as a voltage amplifier. Below a volt, however, due to the non-ideal nature of the semiconductor material, the transistor actually functions exponentially. This means that for every increase in base voltage, the current or "openness" of the transistor also increases exponentially. Replacing the drain resistor with this setup allows for a circuit that is much easier to integrate to a standard keyboard which will output linear voltage jumps.



2.f Emmiter follower

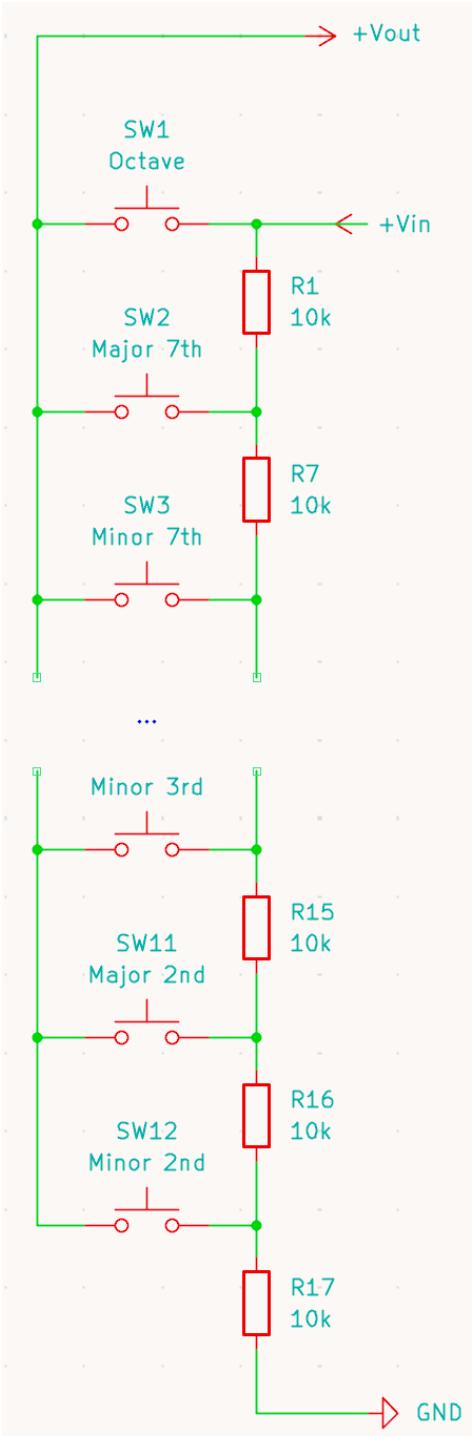
Using an NPN transistor in exponential operating mode is perfect for musical applications but it introduces another issue particular to transistors: their temperature dependence. When operating in an exponential range, NPN transistors open significantly when their ambient temperature increases which for this project's application would mean a constantly changing pitch as the temperature of the room changes. This is not ideal for musical instruments.

To combat this, a PNP transistor is configured in emitter follower configuration with its emitter connected to the base of the PNP transistor. PNP transistors in emitter follower configuration work not unlike a buffer. So when setup as in the diagram below the PNP (Q2) copies Vin to the NPN's (Q1) base node. This is significant because as the temperature increases, the NPN begins to open, but so does the PNP, causing the voltage at the base of the NPN to drop as more voltage can flow out to -12V through the PNP. This proportionally closes the NPN effectively canceling out the temperature dependence by introducing a negative feedback loop. In the circuit, these two transistors should be as close to each other as possible in order to equalize and prevent the tuning from drifting.



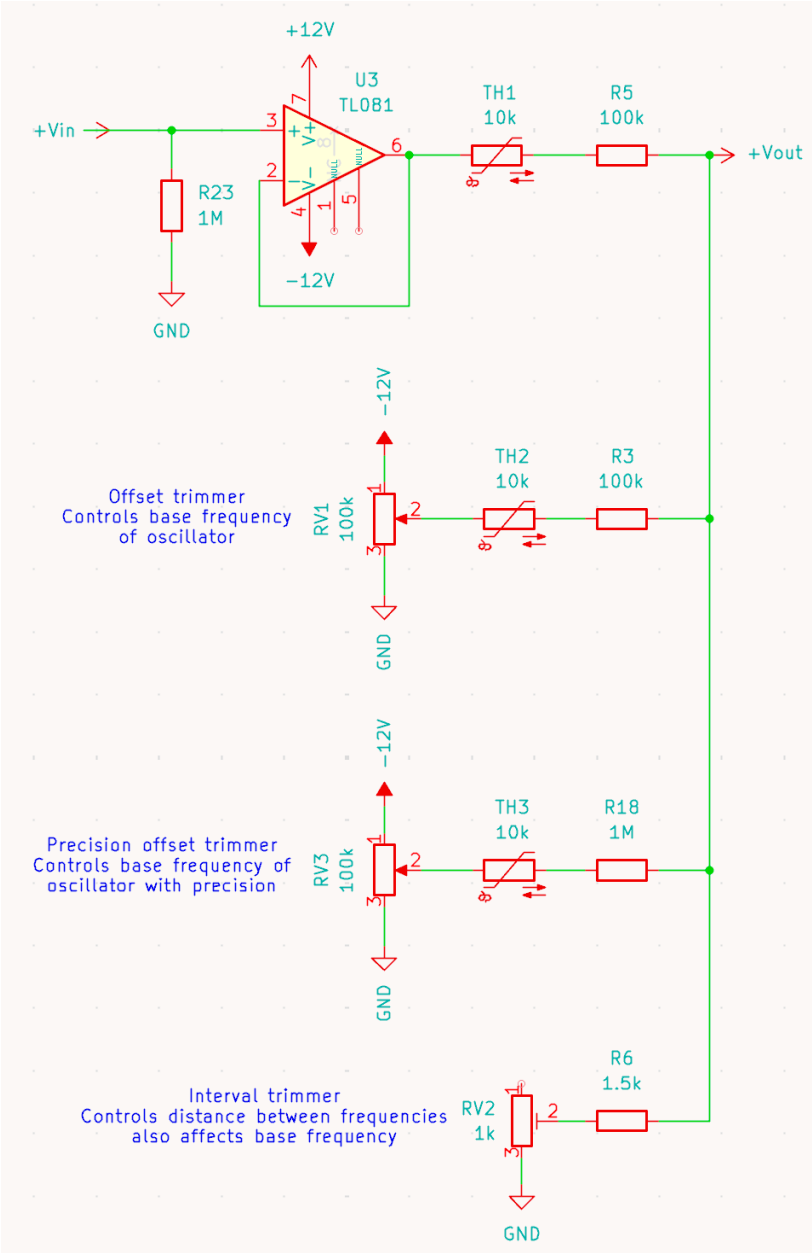
2.g Keyboard

Now the circuit needs a way to vary the voltage being sent to the PNP resistor from the previous section. The first step of this component is the keyboard. This design consists of a 12 step voltage divider. A voltage is sent in at the top and at each resistor junction that voltage is diminished by -1/12th its value and a button is able to send that stepped value to a bus for further processing.



2.h Buffer, Tuner

The output of the keyboard is sent through a TL081 op amp (U3) in buffer configuration and then into a voltage adder. The 1MOhm resistor (R23) is to remove any noise from the keyboard's output.



The first two potentiometers are used to pull down the voltage by adding a negative voltage to the output. The first (RV1) is coarse and the second (RV3) is precise (10x). The bottom potentiometer (RV2) is a precision trimmer and acts as a voltage divider. It is used to accurately tune the distance between notes. The voltages are added from the top three stages, divided by the bottom stage, and then sent into the base of the PNP transistor which opens, opening the NPN transistor, and adjusting the frequency of the oscillator.

PNP transistors open at a slightly faster rate than NPN transistors which can introduce detuning issues as the ambient temperature rises. To combat this, thermistors are used on each summing stage of the tuner to introduce a temperature dependent voltage change and prevent the PNP from lowering the pitch.

3.a Implementation

Design and fabrication

The building process for this project went incredibly smoothly. Most of the components, save for the ICs, were readily available in laboratory supplies so we were able to begin assembly much earlier than other groups. We built it much in the same way that Section 2 outlined its function. We started with the oscillator core, built the square wave converter, and then went back to complete the tuning circuit before finally developing the keyboard system. Many of the components of the oscillator design were inspired by Moritz Klein's VCO series on YouTube.[1]

We made some modifications to the tuning system and the amplification system. Namely we decided to rearrange the tuning circuit so that it would all output neatly in one stage rather than through a series of buffers. This significantly reduced the footprint of the project and simplified the circuitry and debugging process. We also did not have a second amplification stage in the waveform converter component of the oscillator as it was an unnecessary safety measure given the tolerances and allowances of our components.

The keyboard design was our original idea. The op amp buffer in that component of the circuit was particularly difficult to work with as we initially made the erroneous decision to power it with the input voltage. This led to some unpredicted short circuit events that took us an extended period to debug.

In practice we powered both op amps and the schmitt trigger inverter ICs with +-12V and GND. We used the bench power supply from the lab but a more portable version was also achieved using two 9V batteries in series with a common ground and relative +-9V. We used oscilloscopes extensively to debug and test our design. We used Sony headphones to listen to the output.

[1] "DIY VCO Series" *YouTube*, Moritz Klein https://www.youtube.com/watch?v=QBatvo8bCa4

Dialing in resistor and capacitor values was an ongoing process in the design of this circuit. Capacitor C1 was crucial as if it had too large a value it would take too long to charge and turn the clean sawtooth wave into a triangle wave. After much experimentation a value of 2.2nF was determined to be the best fit. The resistor to ground before the second schmitt trigger inverter (U1D) was important in preventing the square waveform from distorting with increased input voltage. The value was decided upon through extended testing and experimentation until the value no longer allowed for distortion. Due to the nature of music, we were unable to use formulas to attain the most optimal component values. Thus testing and checking was the only viable approach. One iteration method which significantly expedited the fabrication process was using a potentiometer to sweep resistance values and then measure its state and swapping in a static resistor at that juncture.

Limitations

In an ideal world, the input voltage for the keyboard would have been 1V. This is due to the eurorack 1V/octave. Unfortunately, due to limitations of the voltage divider and op amp, we were not able to tune the oscillator to a western scale unless we used an input voltage of at least 2V. This is likely due to power loss to ground in our buffers and noise removal stages. With 2V input the tuning of the scale was very pleasant and consistent (within +-5 cents of perfect tune).

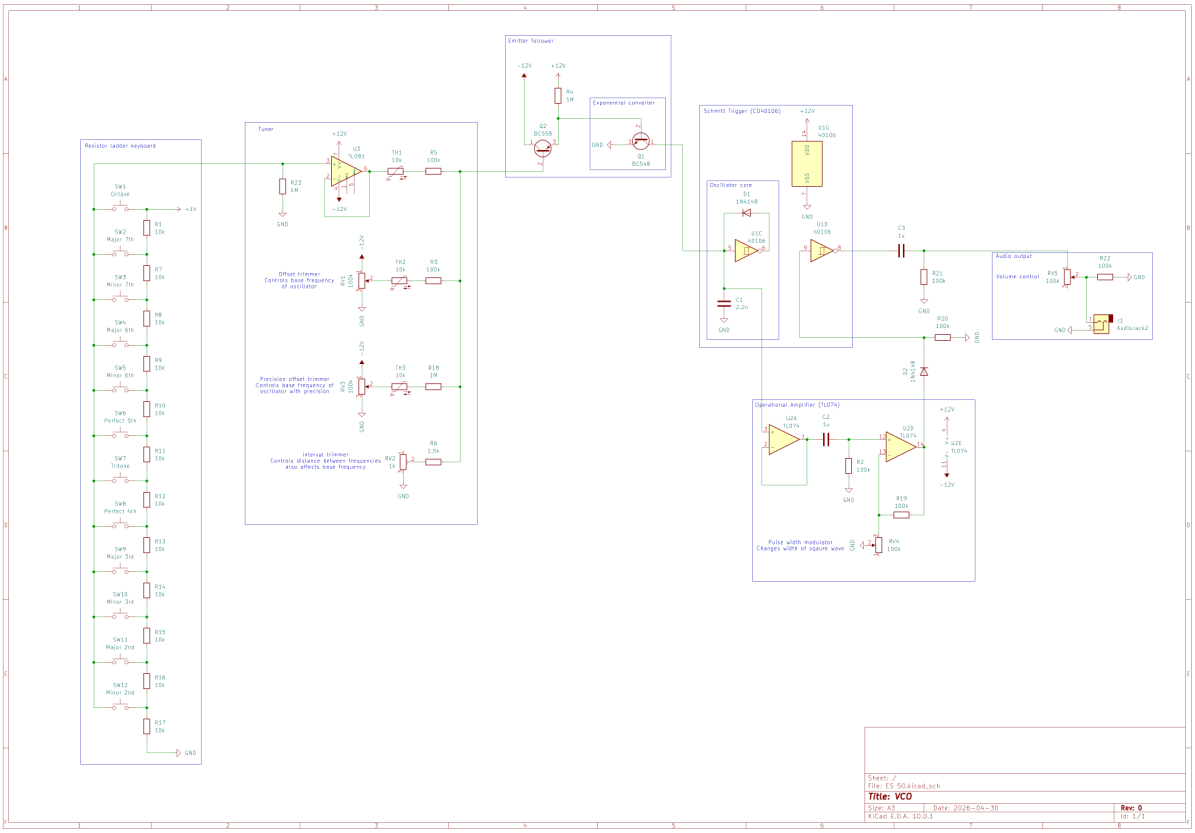
Tuning itself was also a limitation as, even with all our temperature dependent circuitry, the tuning would drift and it took considerable skill and time to retune the circuit. This is likely due to the fact that this was a prototyped project. If we were to put this on a PCB and use components with even tighter tolerances, we are confident that the synthesizer would be more consistent.

Lastly, we intended in our proposal to add filters to the synthesizer allowing even further customization. Unfortunately, though we tried to add a passive low pass filter we ran out of

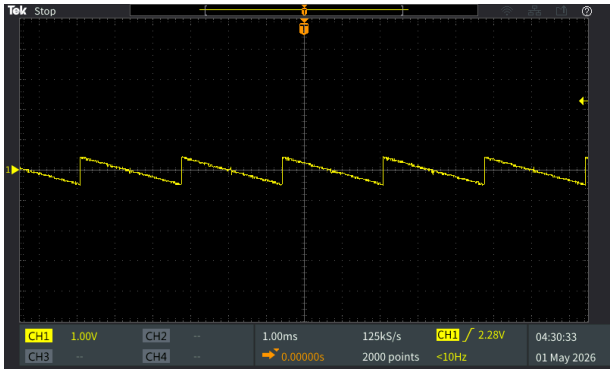
time and had to present without it at the design fair. We intend to develop the synthesizer even further and add such functionality in future.

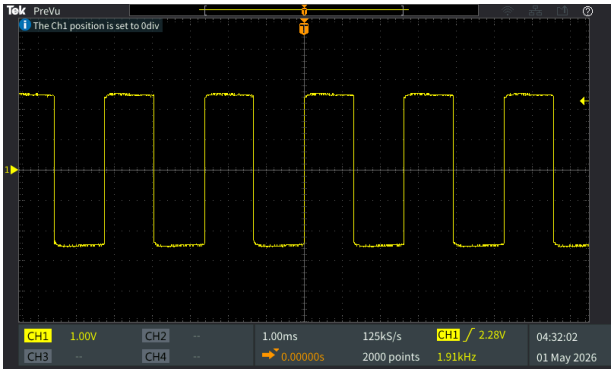
Appendix

Full schematic

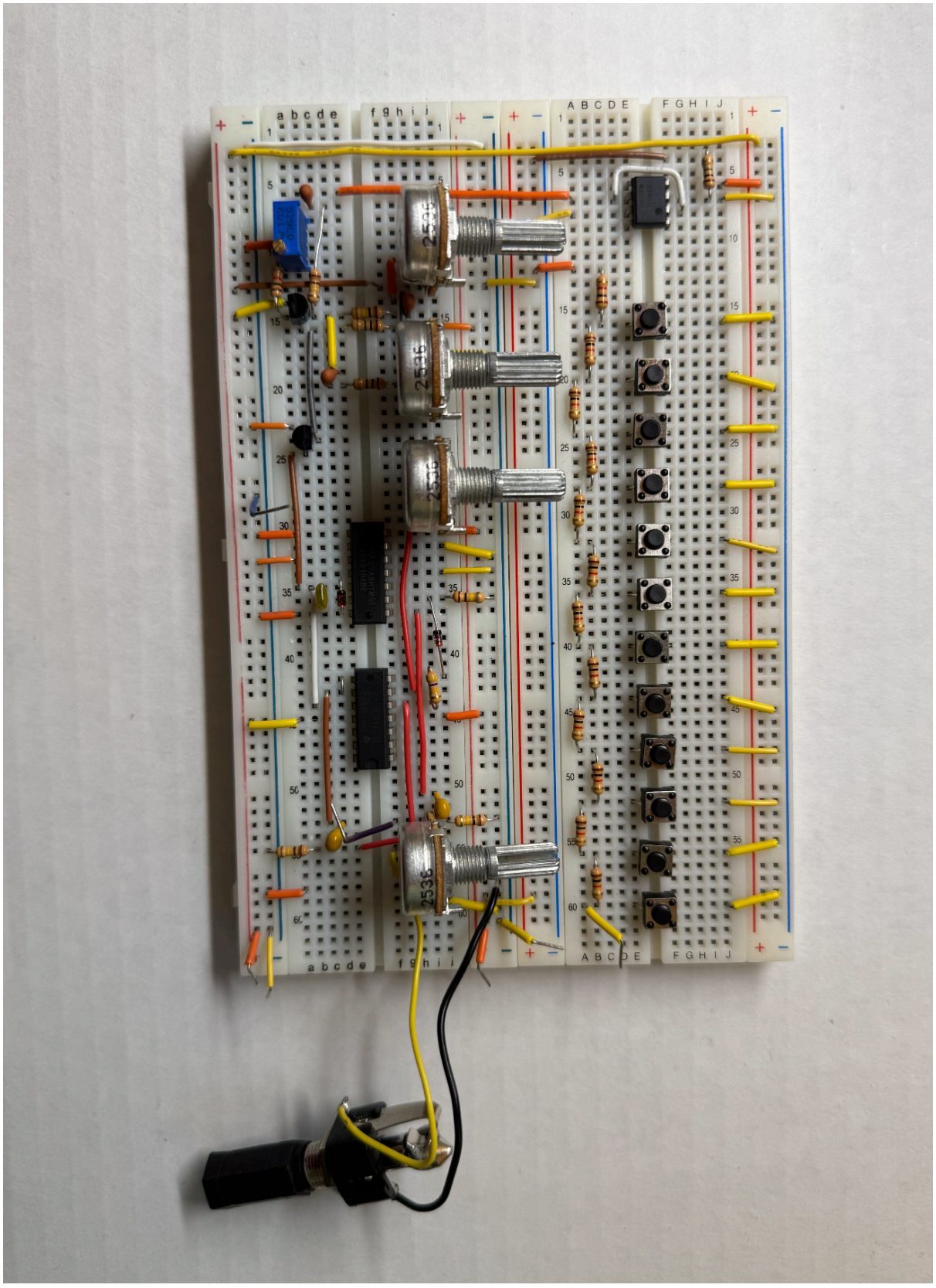


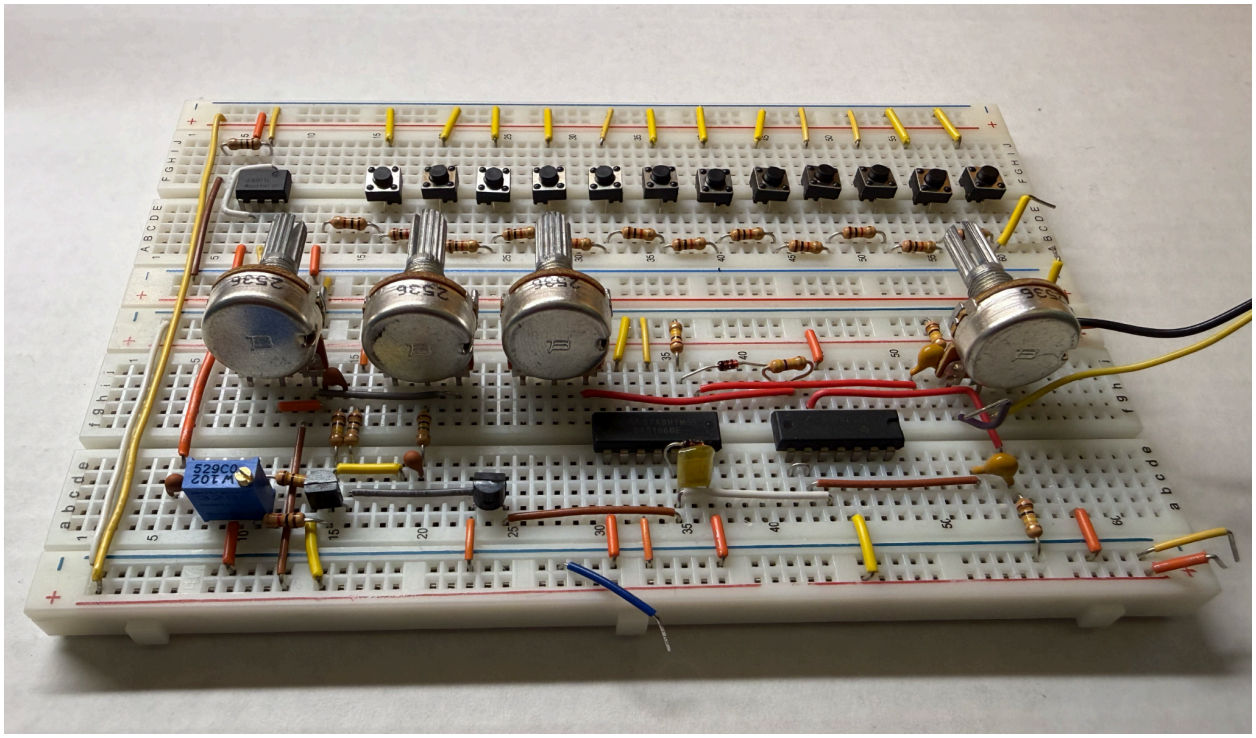
Waveform outputs



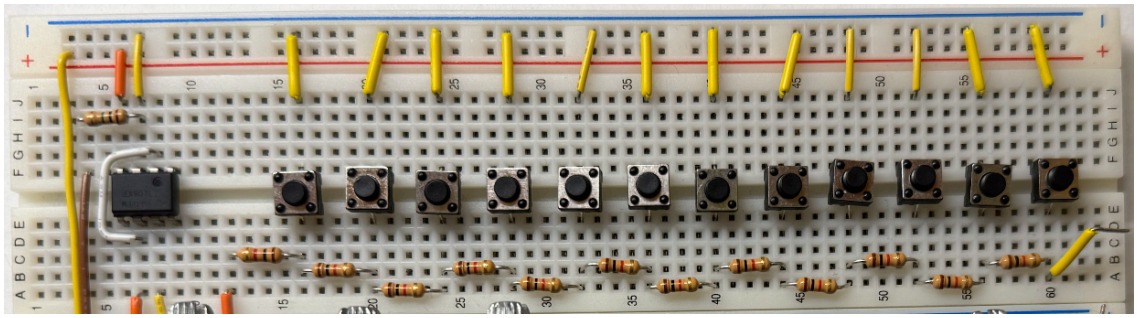


Full circuit



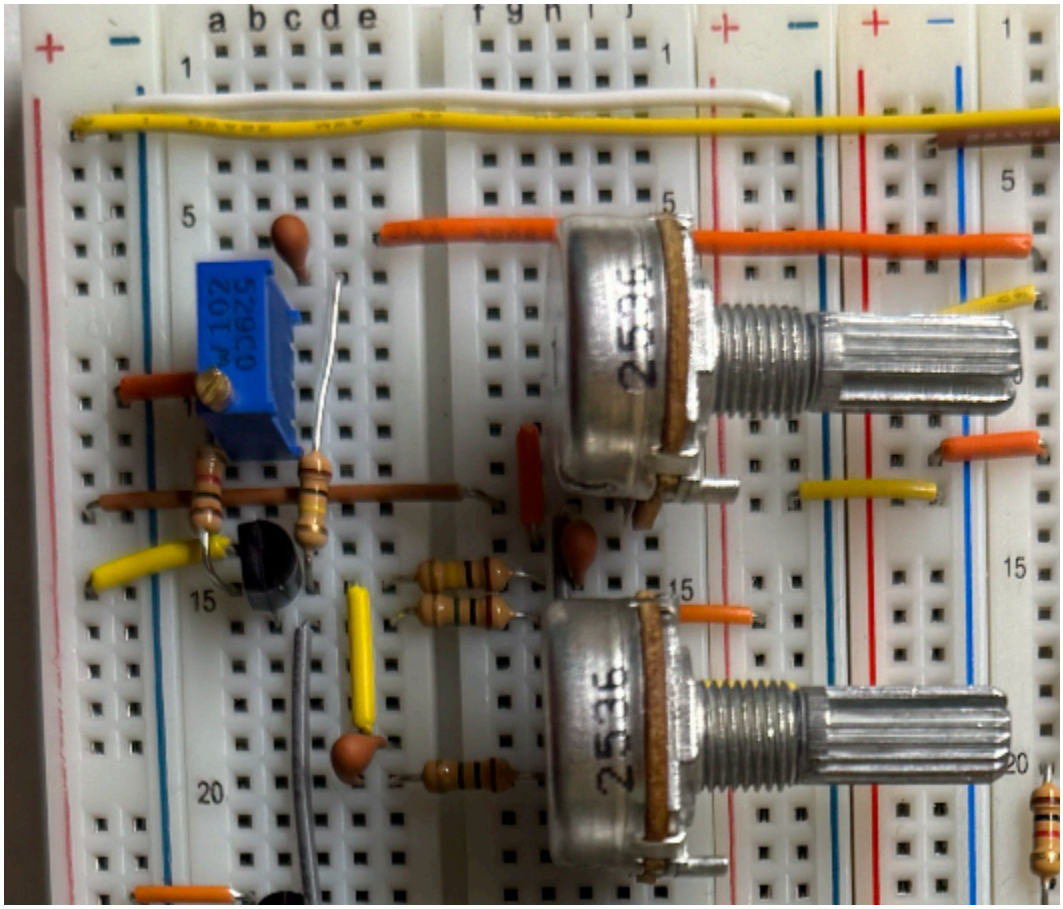


Resistor ladder and associated buffer

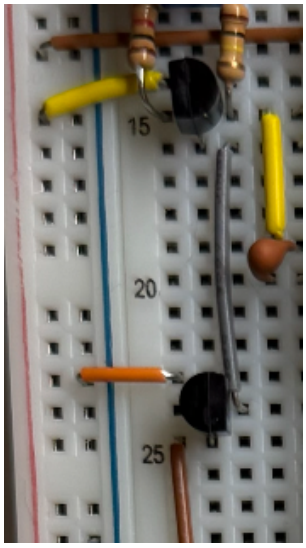


Tuning circuit

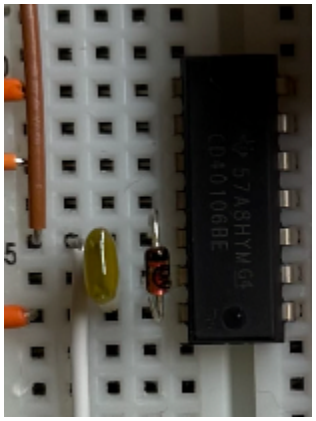
Blue precision trimmer. Top potentiometer course tuning. Bottom potentiometer fine tuning.



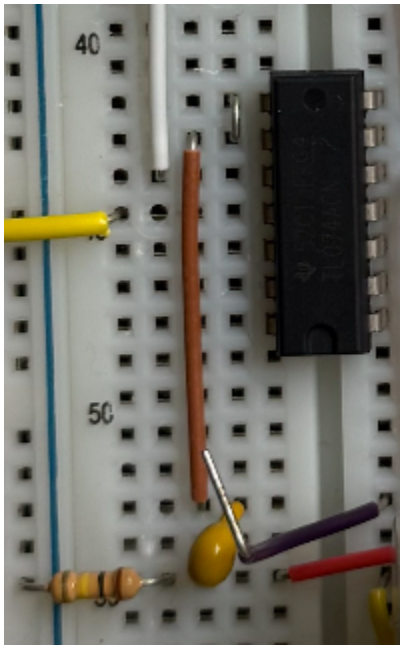
Emitter follower and exponential converter



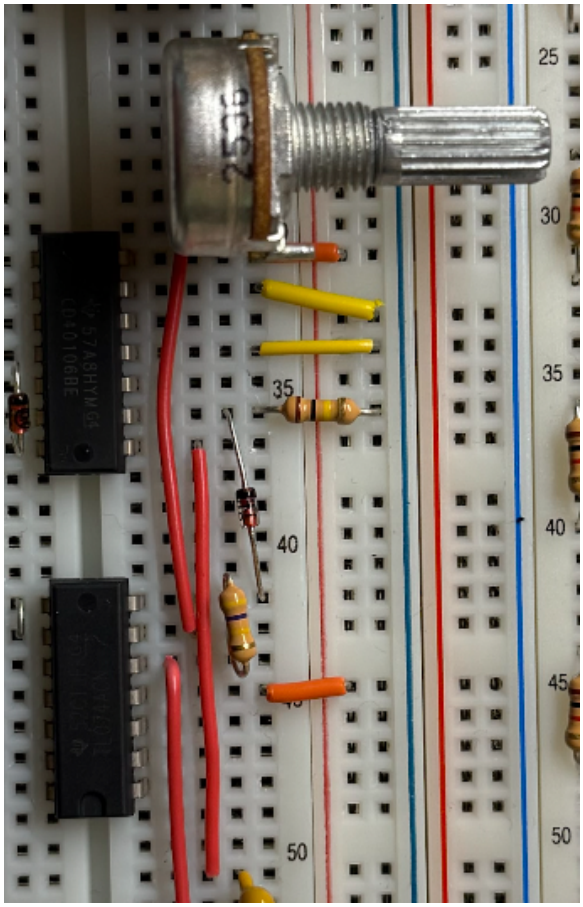
Sawtooth generator



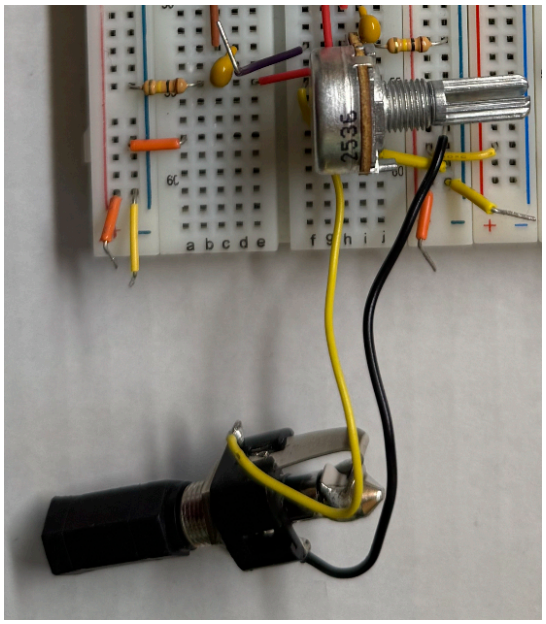
Amplification and AC coupling

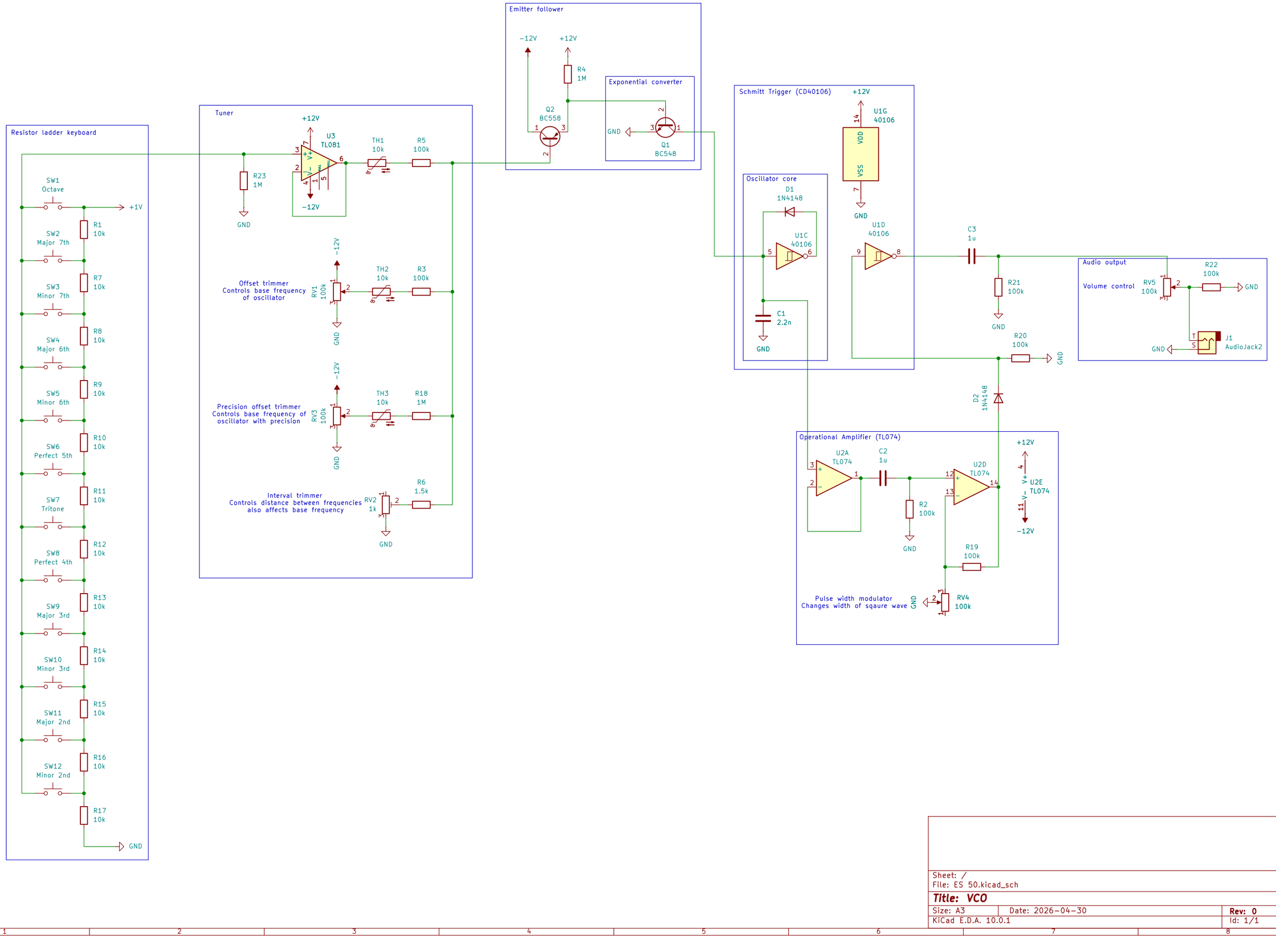


Waveform converter

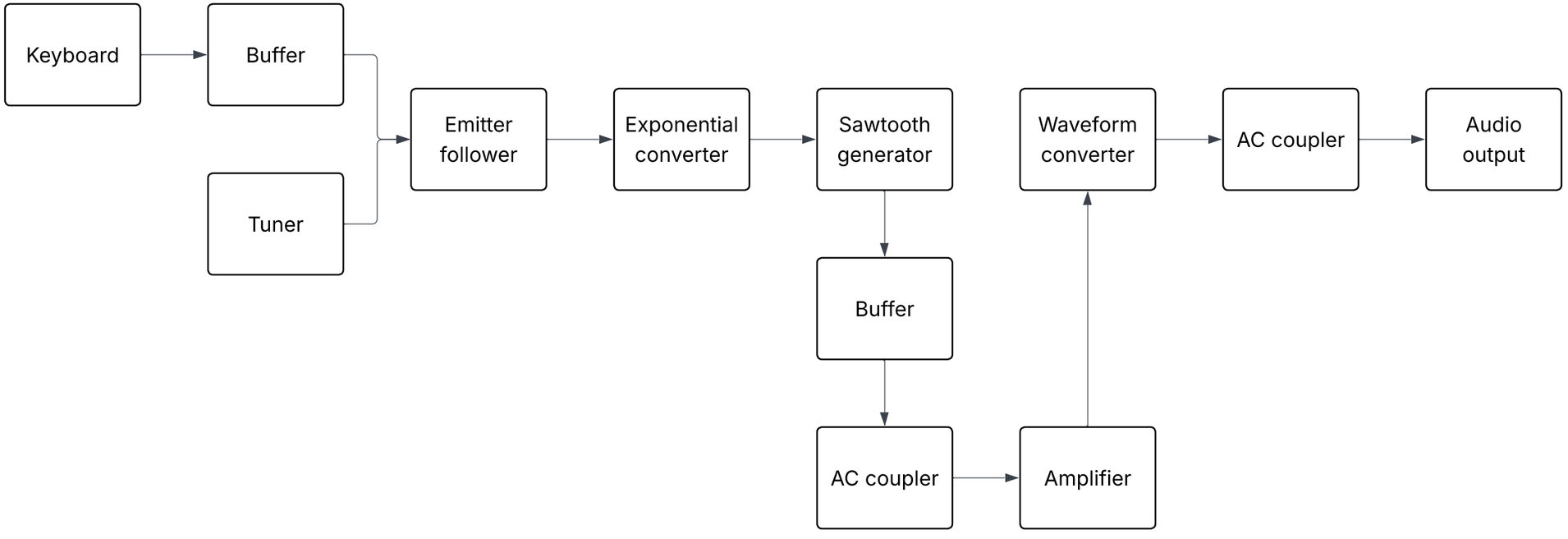


Audio output and AC coupler





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KiCad E.D.A. 10.0.1		Id: 1/1


Keyboard
Buffer
Tuner
Emitter follower
Exponential converter
Sawtooth generator
Buffer
AC coupler
Amplifier
Waveform converter
AC coupler
Audio output

Sources

DIY VCO Series" YouTube, Moritz Klein
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